

OpenTrace RAG Pipeline Architecture

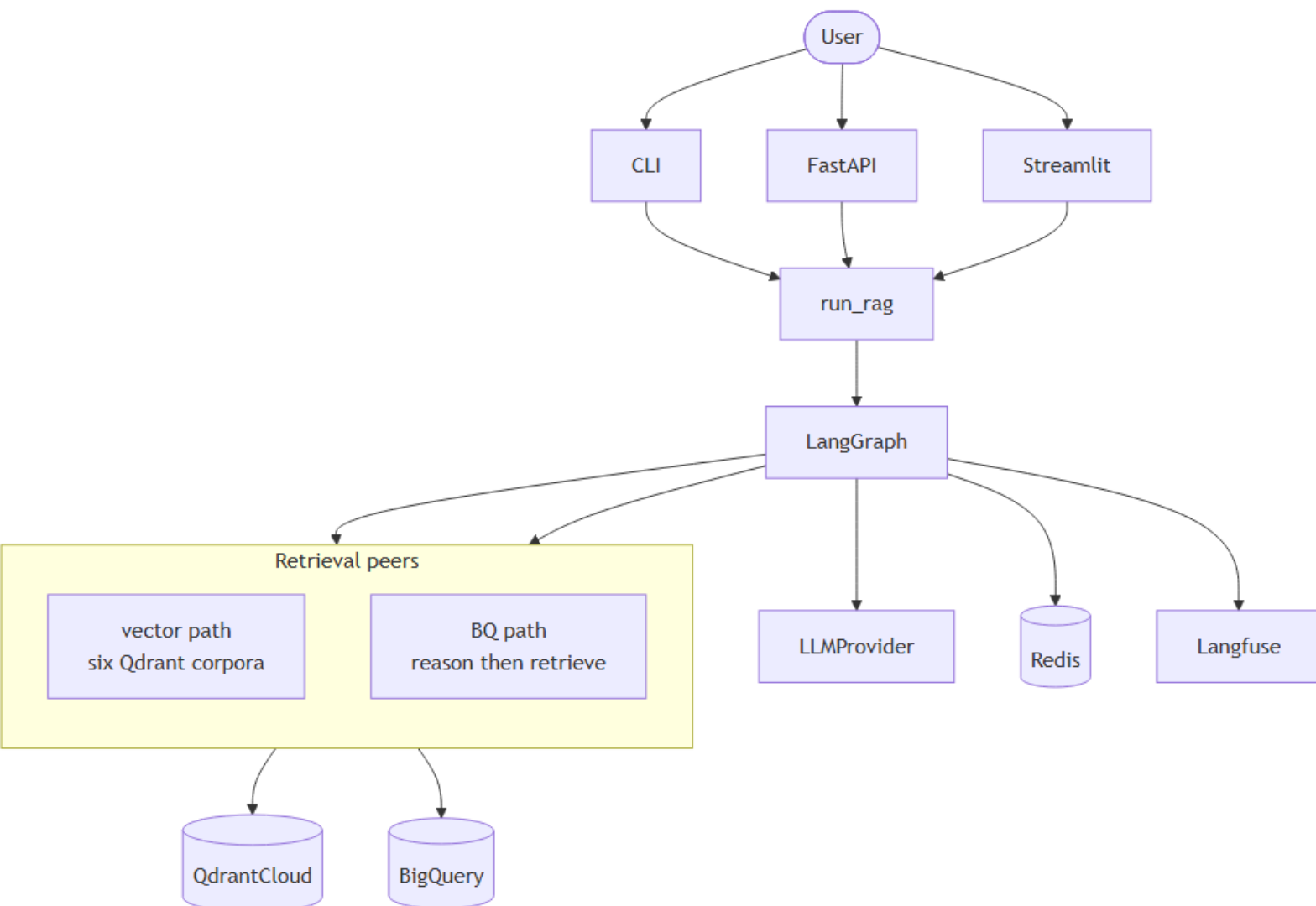
Full LangGraph pipeline, data flows, and node reference
Version 2.0 | Generated 2026-08-15

Purpose and design principles

- Co-equal retrieval: six Qdrant text corpora (unstructured) and BigQuery staging_dev (structured).
- Orchestrate via LangGraph in ml/rag/chatbot/graph.py — vector parallel_retrieve then BQ reason/retrieve, both legs merge.
- Control plane: query enricher (memory) → decompose → Africa panel/ranking → agri measure ontology → task_mode.
- Ontology aids the BQ SQL reasoner (scoped tables/filters); does not replace the LLM; fallback only after retries.
- Retrieval uses the latest user message (enriched if elliptical); chat memory affects generation and enricher.
- BigQuery SQL is staging_dev-only; validated SELECT with LIMIT; Yield/trade/prices/IPC/soil are not Production-locked.
- LLM steps fail soft — empty LLM returns ontology/heuristic fallbacks or context-only answers, not graph crashes.
- ACF (ADZA Confidence Framework) scores cited evidence only (Path B, 0–100) from packed vector + BQ sources.

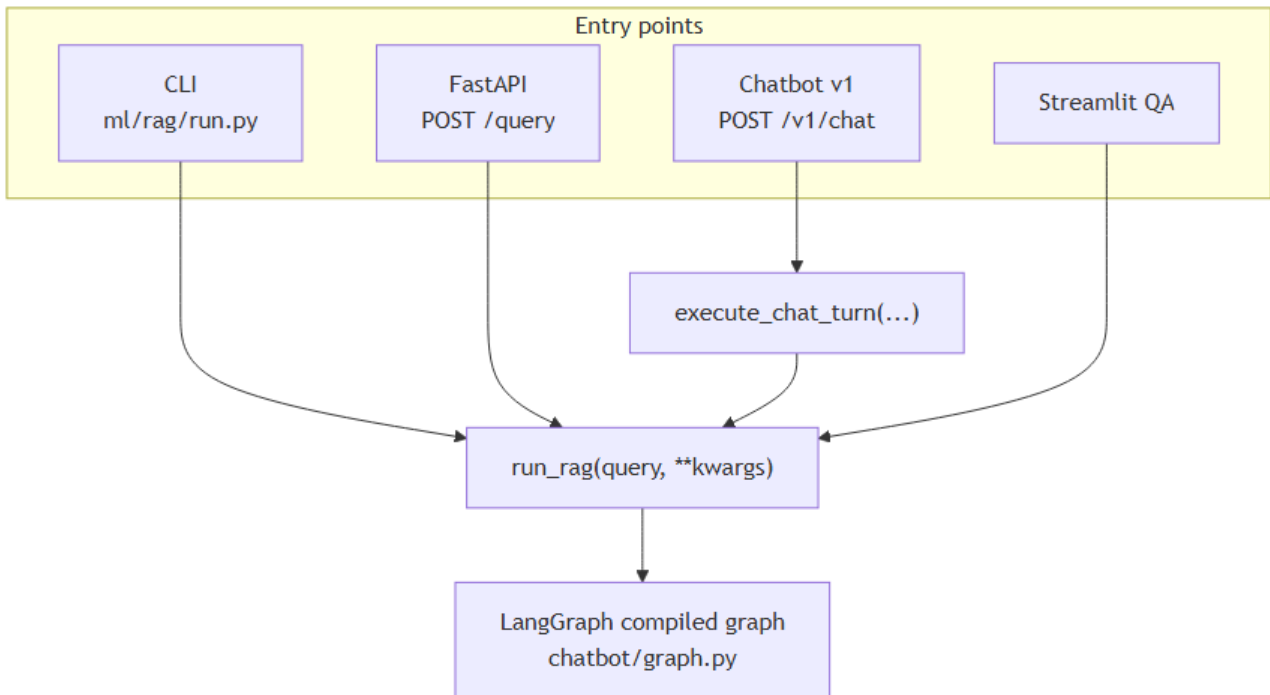
System context

Actors, LangGraph, and external services — Qdrant and BigQuery as peer retrieval backends.



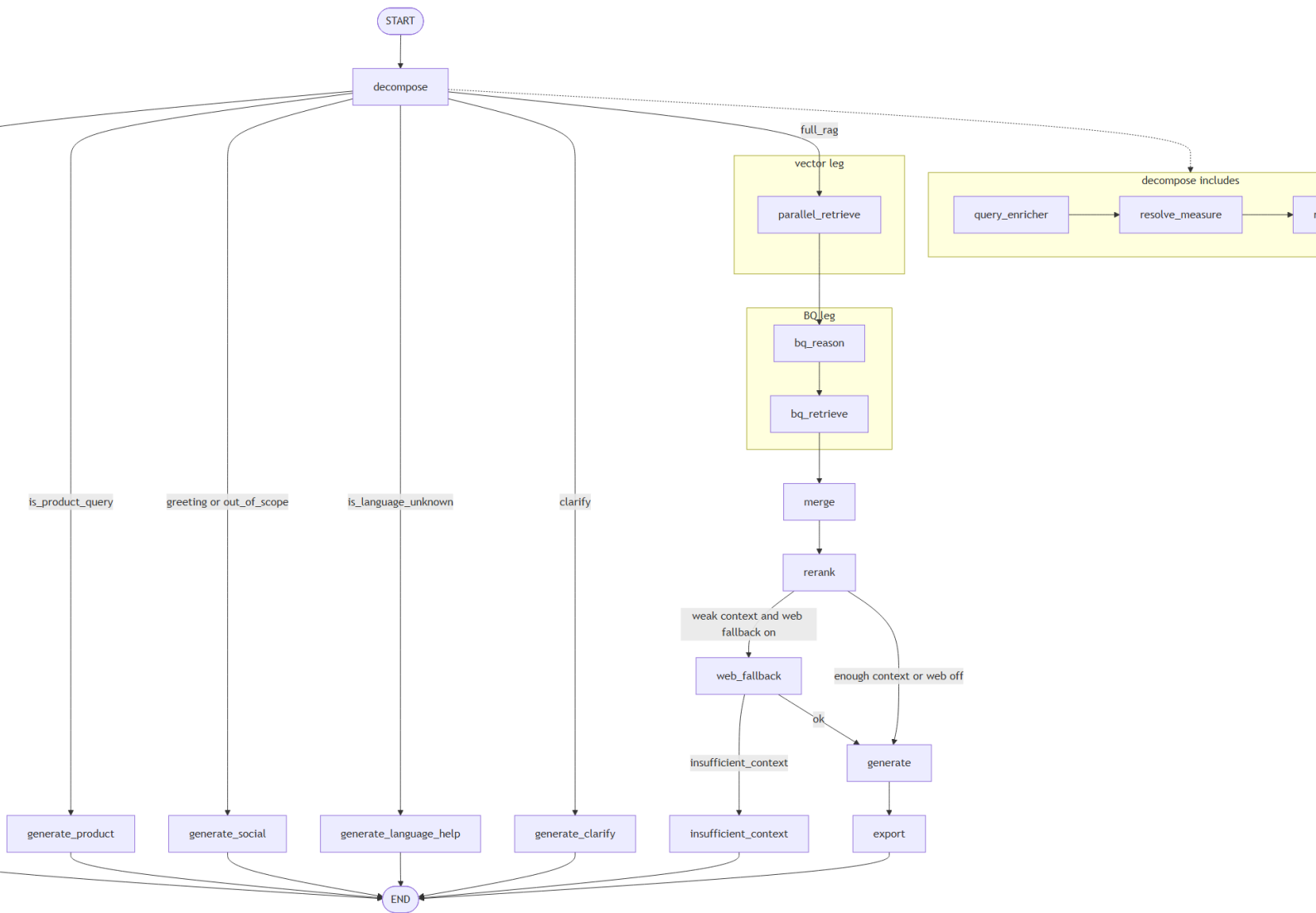
Entry points

CLI, FastAPI, Chatbot v1, and Streamlit routes into run_rag / LangGraph.



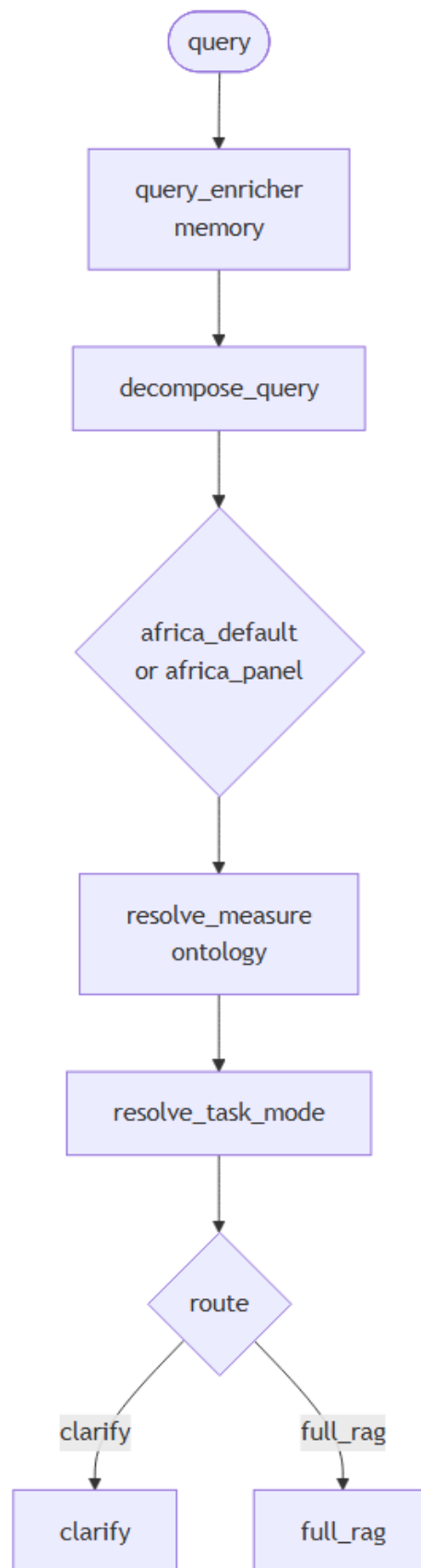
Full LangGraph (E2E)

Start-to-finish graph with vector and BQ retrieval legs into merge.



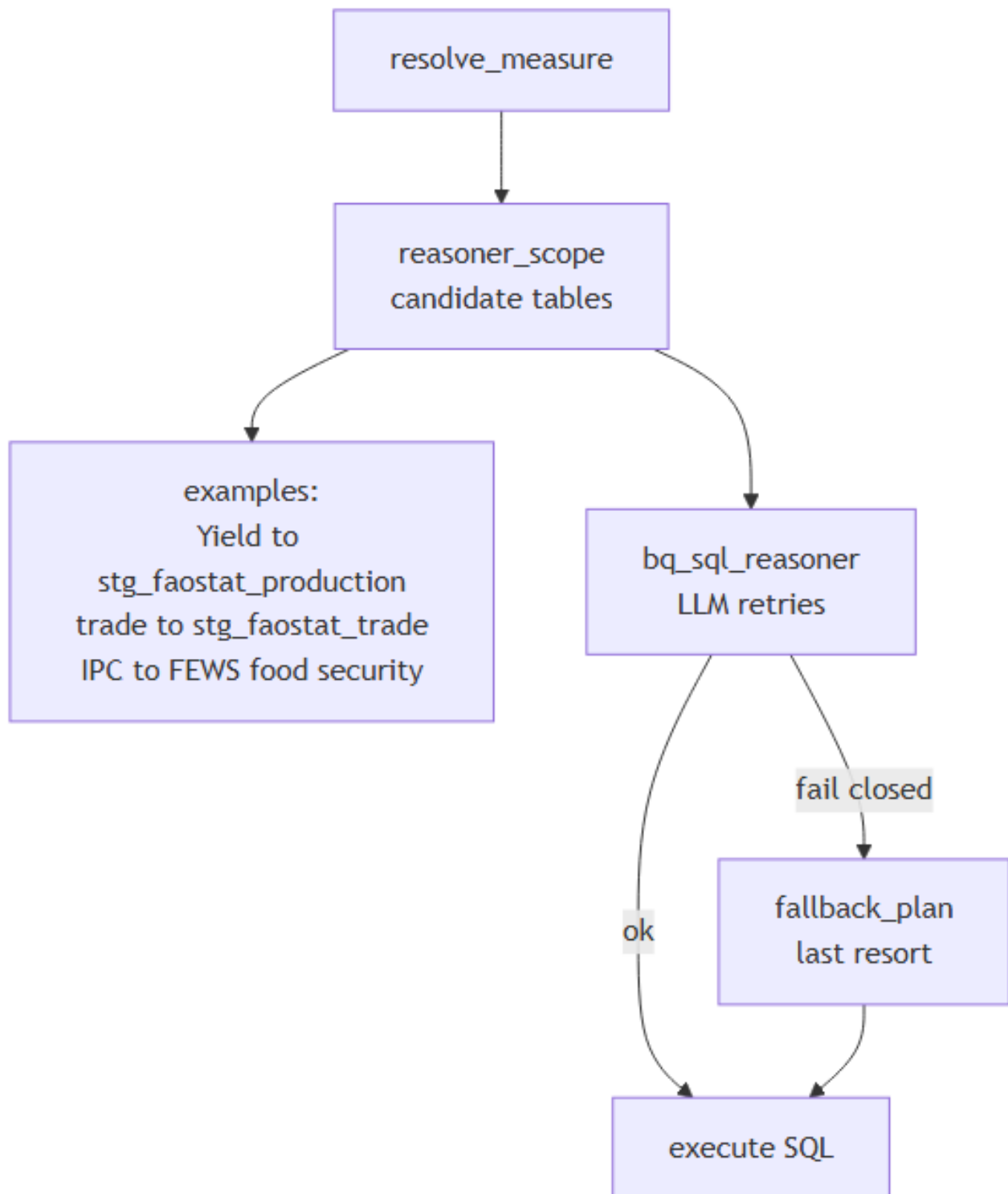
Control plane

Enricher → decompose → Africa scope → measure ontology → task_mode → route.



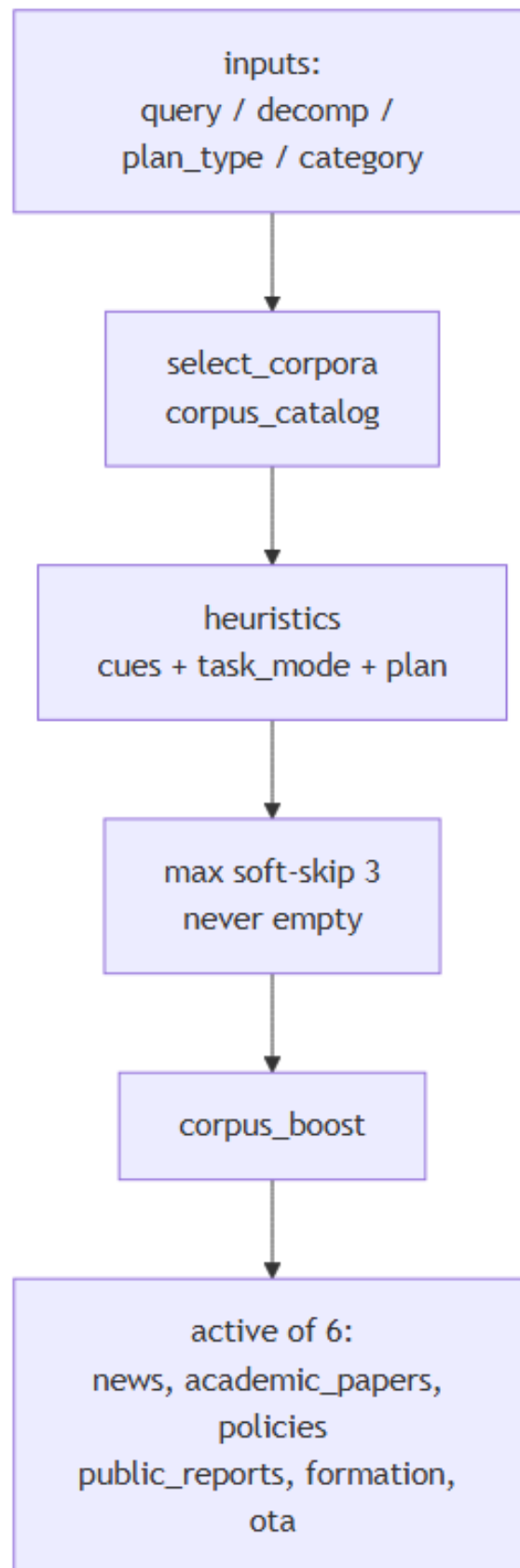
Measure ontology (BQ aid)

resolve_measure → reasoner_scope → LLM reasoner retries → ontology fallback last resort.



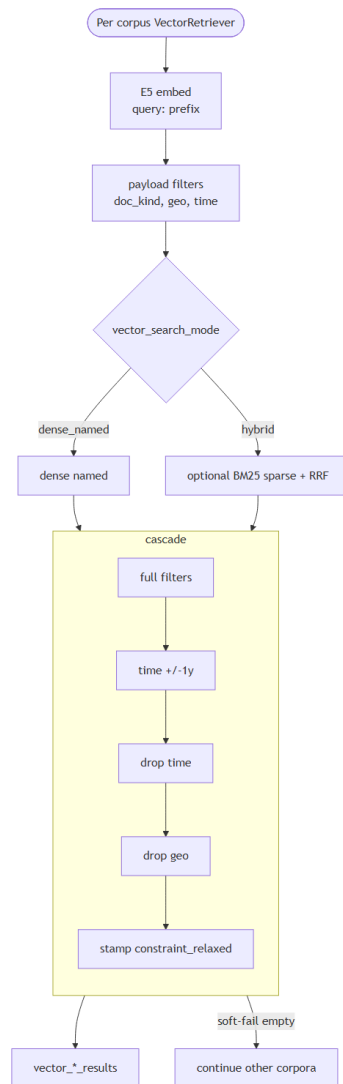
Corpus router

select_corpora gates which of six Qdrant collections to query (max soft-skip 3).



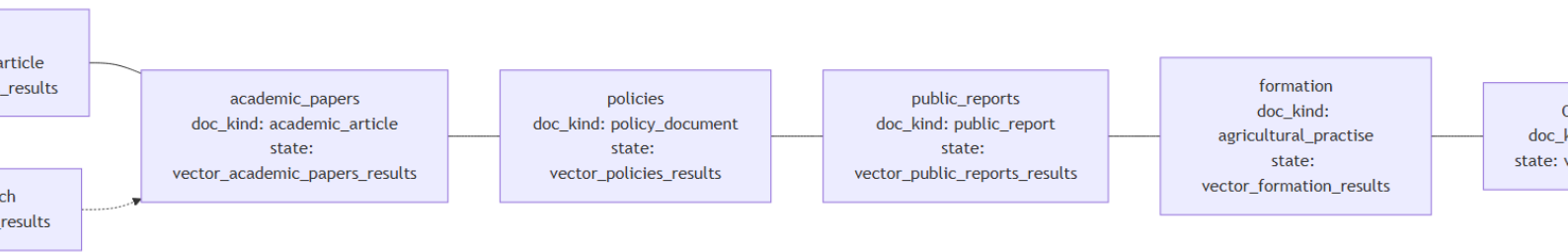
Vector retrieve (per-corpus cascade)

E5 embed → filters → hybrid RRF → filter cascade → vector_*_results.



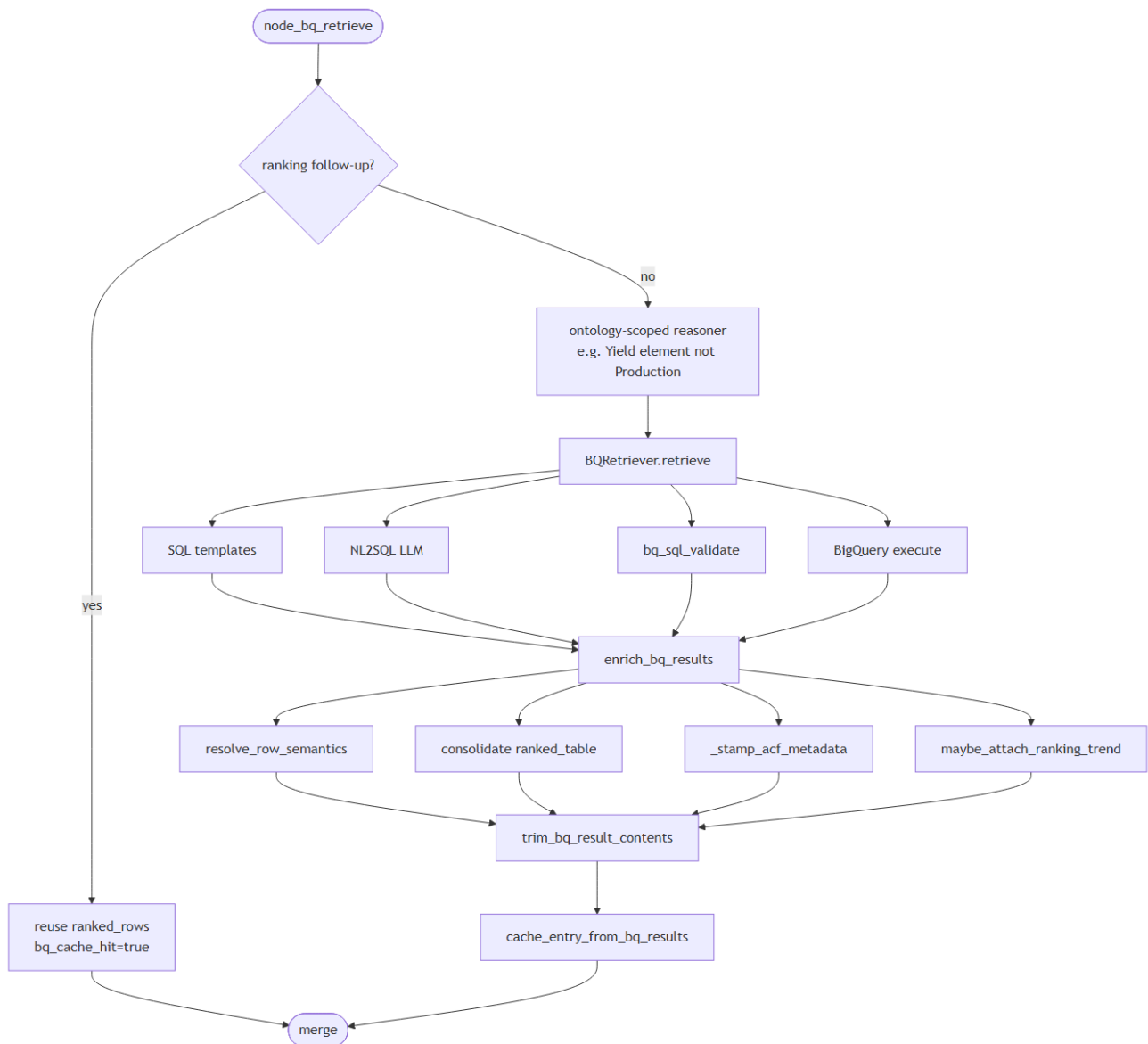
Six Qdrant corpora

Collections, doc_kind, and RAGGraphState vector_* fields.



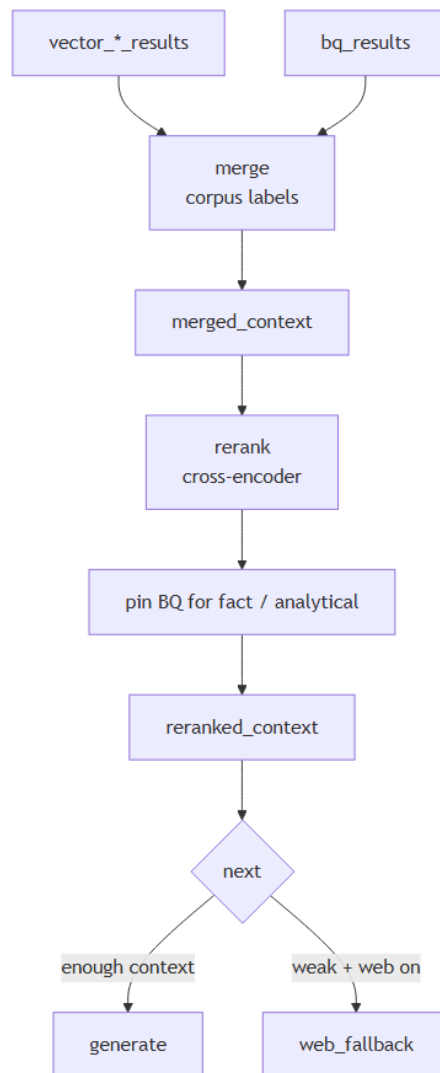
BigQuery retrieve

Ontology-scoped reason → templates/NL2SQL → execute → enrich/cache.



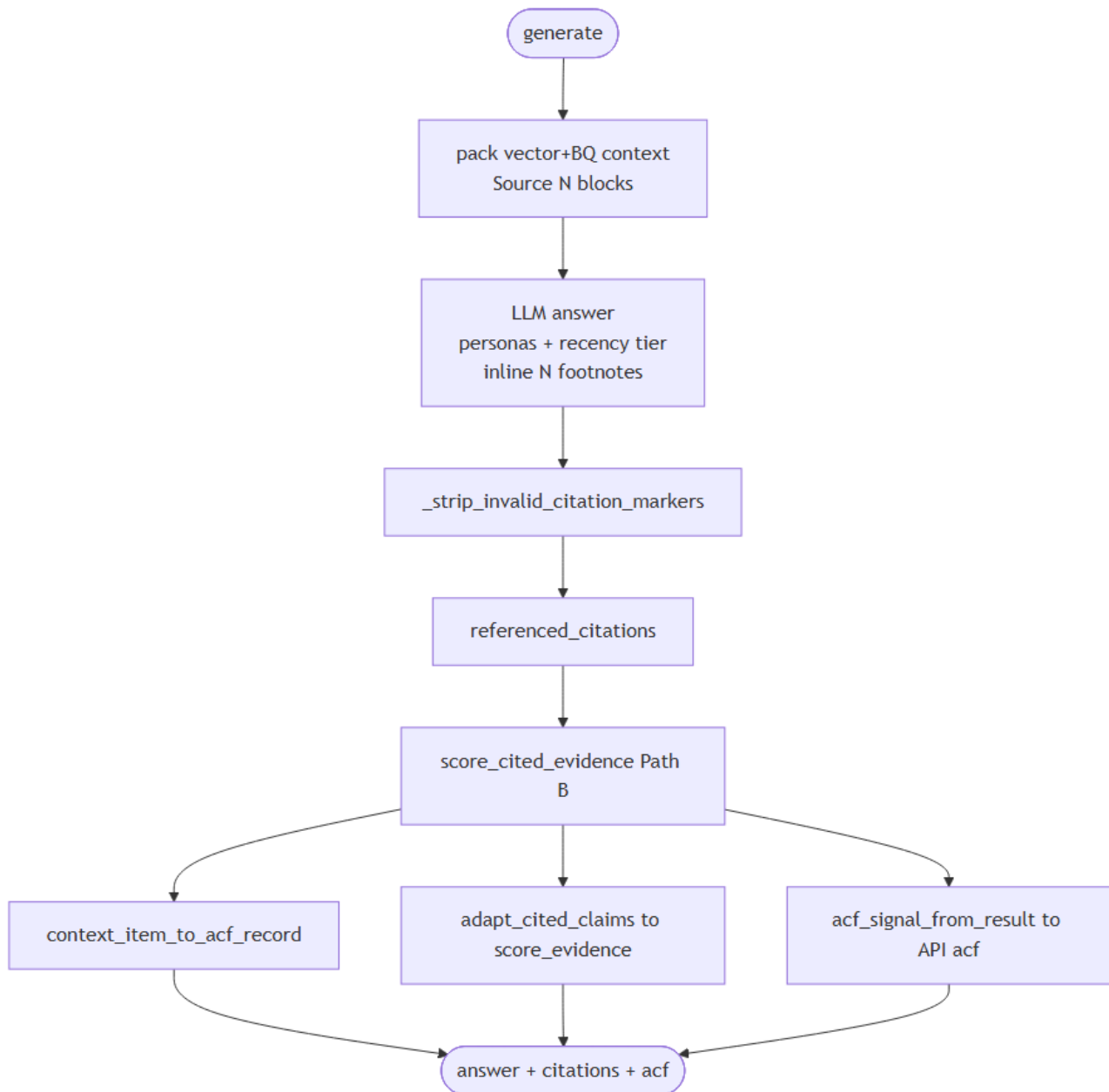
Merge and rerank

Fuse vector chunks + BQ rows; pin BQ for fact/analytical; cross-encoder rerank.



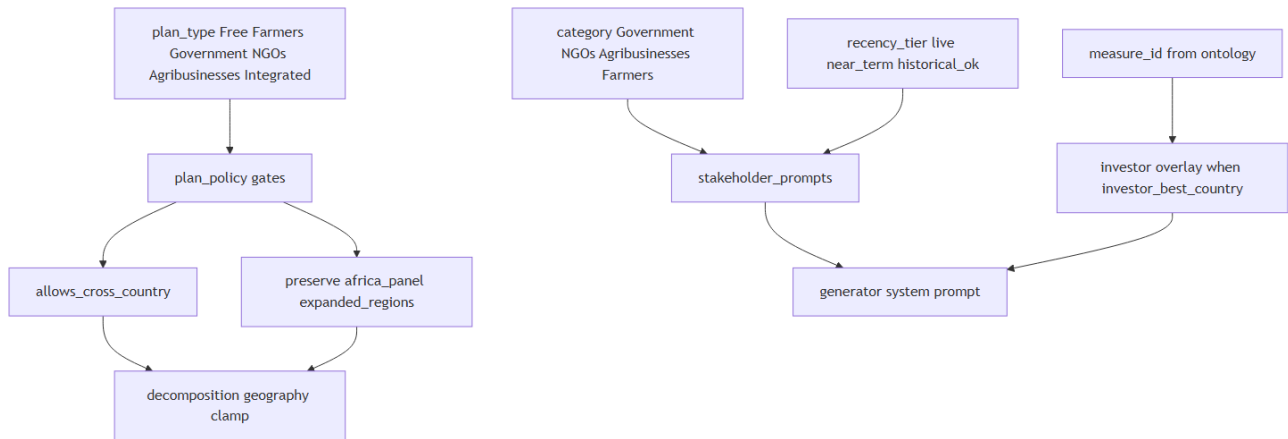
Generation and ACF

Pack vector+BQ context → persona/recency → LLM → citations → ACF Path B.



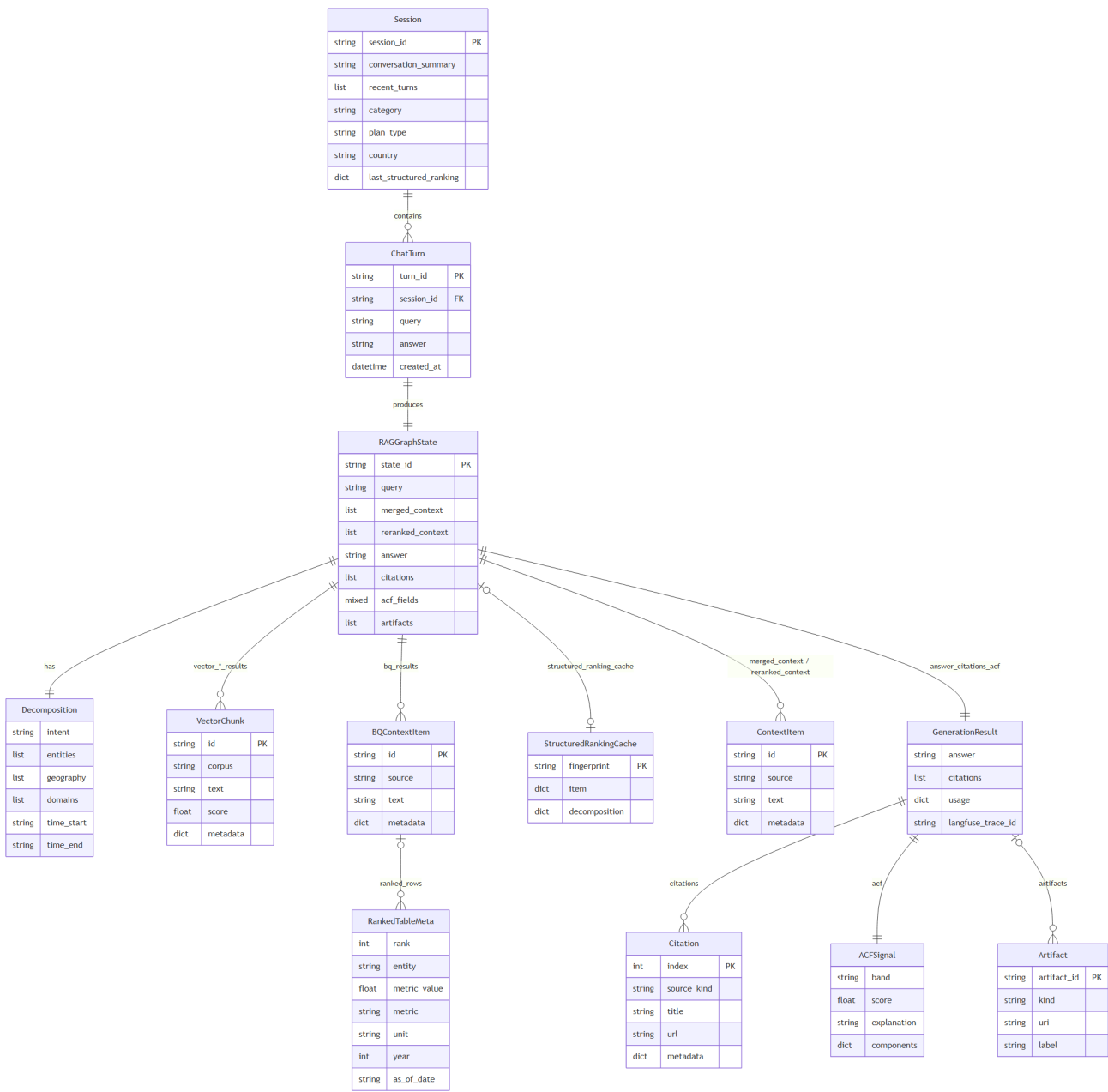
Plan tiers and personas

plan_type gates, category personas, investor overlay, recency tiers.



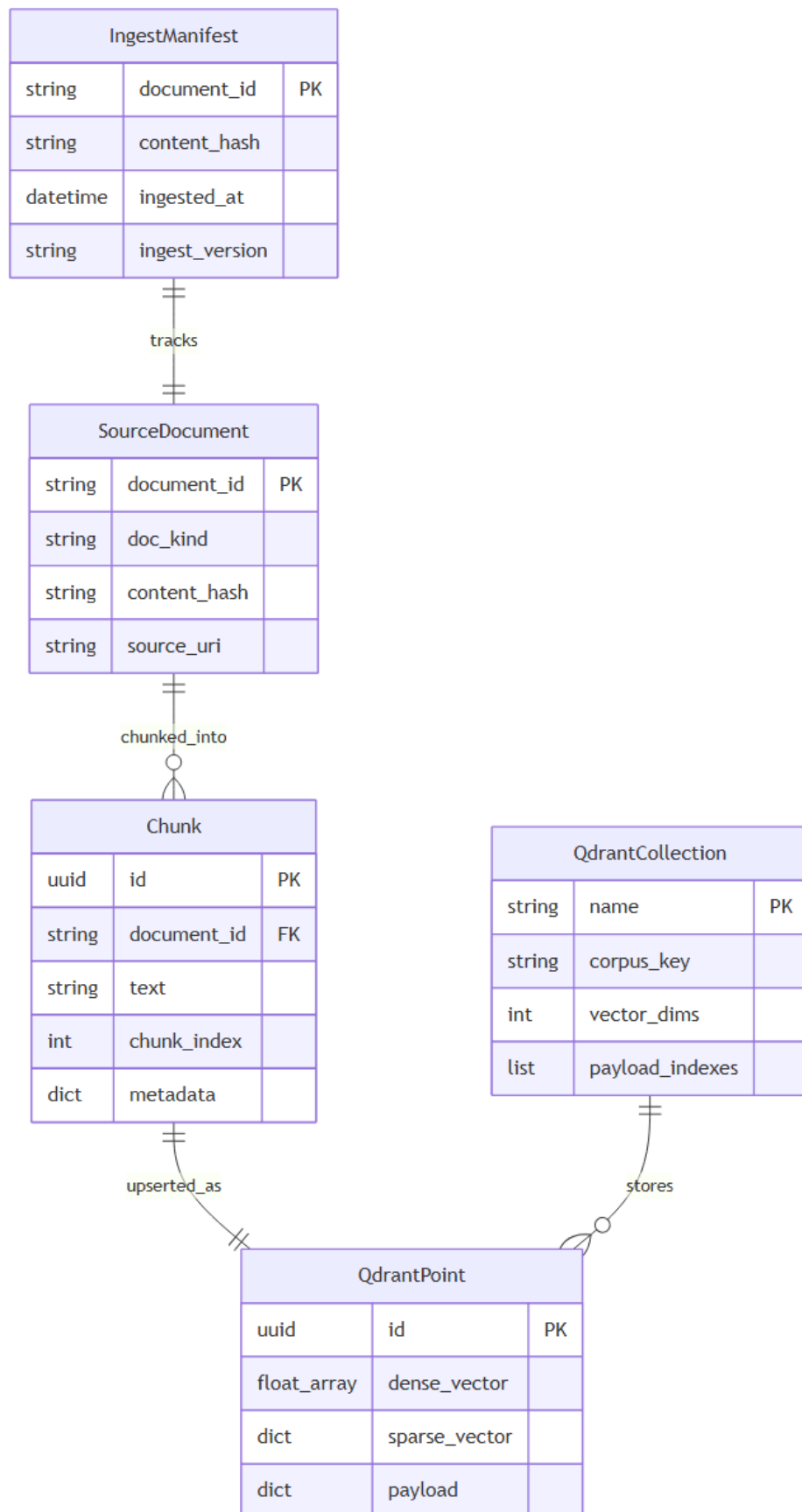
Runtime domain ERD

Query-time session, state, and generation entities.



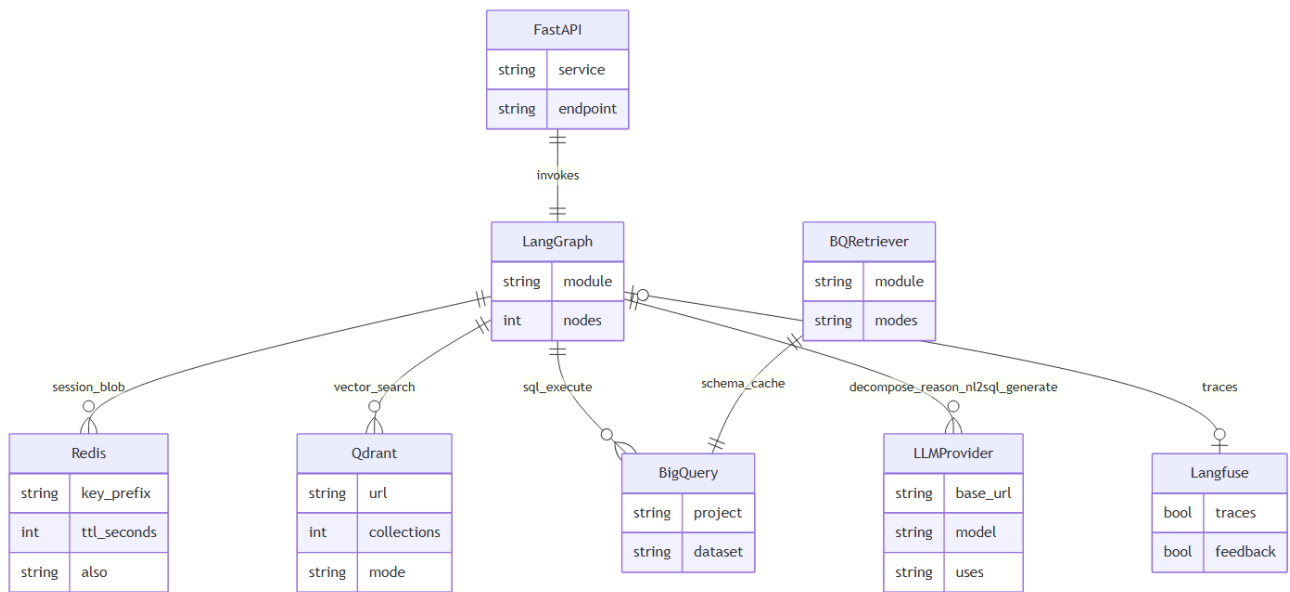
Ingest and vector store ERD

Offline document → chunk → Qdrant pipeline.



Infrastructure ERD

External services invoked by LangGraph.



Node inventory

Node	Type	Description
START	bookend	LangGraph entry; routes to decompose
decompose	routing	Enricher + decompose + Africa scope + ontology + task_mode; short-circuit routes
parallel_retrieve	pipeline	VECTOR LEG: corpus router + concurrent Qdrant retrieve (six corpora)
bq_reason	pipeline	BQ LEG: ontology-scoped table/SQL plan via YAML reasoner + retries
bq_retrieve	pipeline	BQ LEG: execute SQL, enrich results, ranking cache, trend companion
merge	pipeline	Fuse vector_* + bq_results into merged_context
rerank	routing	Cross-encoder rerank; may route to web_fallback or generate
web_fallback	routing	Wikipedia/Tavily append when context weak
insufficient_context	terminal	Deterministic gap message when evidence still insufficient
generate	pipeline	Pack vector+BQ context, persona/recency, LLM, citations, ACF Path B
export	pipeline	CSV/chart/DOCX/PDF artifacts when export_enabled + intent
generate_clarify	terminal	Slot-aware clarify (measure-aware); no retrieval
generate_meta	terminal	Assistant identity / meta answers without retrieval
generate_product	terminal	Product FAQ answers without retrieval
generate_social	terminal	Greetings and out-of-scope polite responses
generate_language_help	terminal	Ask user to rephrase when language unknown
END	bookend	LangGraph exit after terminal or export node

Conditional routing table

Stage	Condition	Target node	Notes
After decompose	task_mode=clarify	generate_clarify	Missing slots per measure ontology
After decompose	is_meta_query	generate_meta	No retrieval; curated ACF
After decompose	is_product_query	generate_product	Product FAQ; curated ACF
After decompose	is_greeting_query OR is_out_of_scope_query	generate_social	Social/gate answer
After decompose	is_language_unknown	generate_language_help	Language help stub
After decompose	else (full_rag)	parallel_retrieve	Vector leg then BQ leg
After parallel_retrieve	always	bq_reason	BQ plan (ontology-scoped)
After bq_reason	always	bq_retrieve	Execute or skip_bq empty
After bq_retrieve	always	merge	Fuse vector + BQ
After merge	always	rerank	Score and trim
After rerank	Usable chunks low + web fallback enabled	web_fallback	Wikipedia/Tavily append
After rerank	Enough context OR web off	generate	LLM generation
After web_fallback	insufficient_context=true	insufficient_context	Deterministic gap message
After web_fallback	else	generate	Proceed with augmented context
After generate	always	export	Artifacts when export_enabled + intent

Control plane notes

- query_enricher merges elliptical/anaphoric follow-ups with prior user turns before decompose.
- decompose_query + africa_default (which-country rank) or africa_panel (full ~54-country panel).
- resolve_measure → task_mode: clarify | analytical | data_export_only | fact_lookup | research | briefing | chat.
- needs_clarify is measure-aware (crop not required for IPC/soil/climate/GDP/investor_best_country).
- plan_policy preserves multi-geo when africa_default / africa_panel / expanded_regions (no Benin amputation).

Runtime domain — entity tables

Session

Field	Type	Key
session_id	str	PK
conversation_summary	str null	
recent_turns	list null	
category	str null	
plan_type	str null	
country	str null	
last_structured_ranking	dict null	Ranking cache blob

ChatTurn

Field	Type	Key
turn_id	str	PK
session_id	str	FK → Session
query	str	
answer	str	
created_at	datetime	

RAGGraphState

Field	Type	Key
state_id	str	PK (implicit per turn)
query	str	
merged_context	list	
reranked_context	list	
answer	str	
citations	list	
acf_*	mixed	ACF signal fields
artifacts	list	

Decomposition

Field	Type	Key
intent	str	
entities	list	
geography	list	
domains	list	
time_start	str null	
time_end	str null	
africa_default	bool null	Which-country Africa ranking
africa_panel	bool null	Full ~54-country panel
expanded_regions	list null	Region keys expanded to countries

VectorChunk

Field	Type	Key
id	str	PK
corpus	str	news academic policy ...
text	str	
score	float	
metadata	dict	geo, published_at, doc_kind

BQContextItem

Field	Type	Key
id	str	PK
source	str	bigquery
text	str	Row prose or ranked_table summary
metadata	dict	sql, bq_enrichment, ranked_rows

StructuredRankingCache

Field	Type	Key
fingerprint	str	PK
item	dict	Cached ranked_rows context item
decomposition	dict	Matching decomposition snapshot

ContextItem

Field	Type	Key
id	str	PK
source	str	vector bigquery web
text	str	
metadata	dict	

GenerationResult

Field	Type	Key
answer	str	
citations	list	
usage	dict	LLM tokens
langfuse_trace_id	str null	

Citation

Field	Type	Key
index	int	PK
source_kind	str	
title	str	
url	str null	
metadata	dict	

ACFSignal

Field	Type	Key
band	str	High Moderate Low
score	int float	0–100
explanation	str	
components	dict null	T/A/F/G breakdown

Artifact

Field	Type	Key
artifact_id	str	PK
kind	str	csv chart docx pdf
uri	str	GCS or local URI
label	str	

RankedTableMeta

Field	Type	Key
rank	int	
entity	str	Country/crop name
metric_value	float	
metric	str	
unit	str	
year	int null	
as_of_date	str	Required for ACF

Relationships

From	Cardinality	To	Label
Session	1:N	ChatTurn	contains
ChatTurn	1:1	RAGGraphState	produces
RAGGraphState	1:1	Decomposition	has
RAGGraphState	1:N	VectorChunk	vector_*_results
RAGGraphState	1:N	BQContextItem	bq_results
RAGGraphState	1:0..1	StructuredRankingCache	structured_ranking_cache
RAGGraphState	1:N	ContextItem	merged_context / reranked_context
RAGGraphState	1:1	GenerationResult	answer_citations_acf
GenerationResult	1:N	Citation	citations
GenerationResult	1:1	ACFSignal	acf
GenerationResult	0:N	Artifact	artifacts
BQContextItem	0:N	RankedTableMeta	ranked_rows

Ingest and vector store — entity tables

SourceDocument

Field	Type	Key
document_id	str	PK
doc_kind	str	news_article academic_article ...
content_hash	str	SHA for incremental skip
source_uri	str null	Drive path or file path

Chunk

Field	Type	Key
id	UUID5	PK
document_id	str	FK → SourceDocument
text	str	
chunk_index	int	
metadata	dict	geo, dates, lanes

QdrantPoint

Field	Type	Key
id	UUID5	PK (= Chunk.id)
dense_vector	float[]	
sparse_vector	dict null	BM25 sparse when hybrid
payload	dict	doc_kind, country, published_at, ...

QdrantCollection

Field	Type	Key
name	str	PK
corpus_key	str	news academic_papers ...
vector_dims	int	
payload_indexes	list	geo, dates, doc_kind

IngestManifest

Field	Type	Key
document_id	str	PK/FK
content_hash	str	
ingested_at	datetime	
ingest_version	str	Bump to force rebuild

Relationships

From	Cardinality	To	Label
SourceDocument	1:N	Chunk	chunked_into

From	Cardinality	To	Label
Chunk	1:1	QdrantPoint	upserted_as
QdrantCollection	1:N	QdrantPoint	stores
IngestManifest	1:1	SourceDocument	tracks

Infrastructure — entity tables

FastAPI

Field	Type	Key
service	str	RAG API / Chatbot v1 / Streamlit
endpoint	str	POST /query, /v1/chat, ...

LangGraph

Field	Type	Key
module	str	chatbot/graph.py
nodes	16	decompose ... export + terminals

Redis

Field	Type	Key
key_prefix	str	rag:session:*
ttl_seconds	int	RAG_SESSION_TTL_SECONDS
also	str	BQ schema cache

Qdrant

Field	Type	Key
url	str	QDRANT_URL
collections	6	news, academic, policy, ...
mode	str	hybrid dense+sparse (optional)

BigQuery

Field	Type	Key
project	str	BQ_PROJECT
dataset	str	staging_dev (BQ_DATASET_SILVER)

LLMProvider

Field	Type	Key
base_url	str	RAG_LLM_BASE_URL
model	str	RAG_LLM_MODEL_ID
uses	str	decompose, NL2SQL, generate, ...

Langfuse

Field	Type	Key
traces	bool	Optional observed spans per node
feedback	bool	POST /feedback scores

BQRetriever

Field	Type	Key
module	str	bq_retriever.py
modes	str	templates + NL2SQL per_hint batch

Relationships

From	Cardinality	To	Label
FastAPI	1:1	LangGraph	invokes
LangGraph	1:N	Redis	session_blob
LangGraph	1:N	Qdrant	vector_search
LangGraph	1:N	BigQuery	sql_execute
LangGraph	1:N	LLMProvider	decompose_reason_nl2sql_generate
LangGraph	0:1	Langfuse	traces
BQRetriever	1:1	BigQuery	schema_cache

BQ enrich and ACF metadata

- `enrich_bq_results` stamps year, `as_of_date`, metric, unit from SQL/decomposition.
- `ranked_table` consolidation merges multi-row BQ rankings into one context item.
- `maybe_attach_ranking_trend` runs FAOSTAT Production+Yield bracketing years (Y-1 vs Y+1).
- Session ranking cache (`last_structured_ranking`) skips BQ on compatible follow-ups.
- `context_item_to_acf_record` requires `as_of_date` — `ranked_table` must be stamped.

Node reference

Detailed behavior for each LangGraph node.

Node: decompose

Enrich elliptical follow-ups, decompose facets, resolve measure/task_mode, classify route.

Attribute	Value
Module(s)	chatbot/query_enricher.py, query_decomposer.py, agri_measure_ontology.py, task_mode.py, plan_policy.py
Reads state	query, conversation_summary, recent_turns, plan_type, category, user_profile
Writes state	query (if enriched), enriched_query, decomposition, measure_id, recency_tier, task_mode, analytical_mode, route flags, answer_lang, export_intent
LLM	Optional (decomposer enrichment)
External I/O	None
Failure behavior	Heuristics always succeed; LLM failure keeps heuristic decomposition.

Sub-steps

- `enrich_query_with_memory` for anaphora / short follow-ups.
- `decompose_query` + `africa_default` or `africa_panel` annotation.
- `expand_regions_in_decomposition` for analytical/export/panel modes.
- `apply_plan_decomposition_gates` (preserve Africa panels).
- `resolve_measure` + `resolve_task_mode` (clarify / analytical / export / fact / research / briefing / chat).
- Detect meta, product, greeting, out-of-scope, unknown language.

Node: parallel_retrieve

VECTOR LEG — retrieve unstructured context from up to six Qdrant corpora in parallel.

Attribute	Value
Module(s)	chatbot/graph.py, corpus_catalog.py, retrievers/vector_retriever.py, sparse_embeddings.py
Reads state	query, decomposition, corpus_selection inputs, geo/time overrides, top_k, plan_type, task_mode, category
Writes state	vector_*_results, vector_results, corpus_selection
LLM	No
External I/O	Qdrant Cloud
Failure behavior	Empty corpus returns []; graph continues.

Sub-steps

- `select_corpora`: heuristic gate, max soft-skip 3, never empty, stamp `corpus_boost`.
- Thread pool: one `VectorRetriever.retrieve` per selected corpus.
- E5 embed with query: prefix; payload filters `doc_kind`/geo/time/domains.
- `dense_named` or hybrid (BM25 sparse + RRF) per collection profile.
- Cascade: full filters → time $\pm 1y$ → drop time → drop geo; stamp `constraint_relaxed`.

- Geo post-filter; soft-fail empty corpora; continue graph.

Node: bq_reason

BQ LEG planner — ontology-scoped staging_dev table/SQL intents.

Attribute	Value
Module(s)	chatbot/bq_sql_reasoner.py, agri_measure_ontology.py, bq_table_schema_yaml.py, fact_bq_plan.py, analytical_bq_plan.py
Reads state	query, decomposition, plan_type, category, task_mode, analytical_mode
Writes state	bq_sql_plan, bq_table_candidates
LLM	Yes (RAG_BQ_REASONER_MODEL_ID or default)
External I/O	None
Failure behavior	Ontology/heuristic fallback when measure known; else skip_bq empty plan.

Sub-steps

- resolve_measure + reasoner_scope; format_reasoner_index(table_ids=candidates).
- Forced analytical/fact/export plans from MeasureSpec when task_mode matches.
- Else LLM reasoner with retries; ontology fallback_plan or Africa heuristic last resort.
- Emit table_hints packs for NL2SQL.

Node: bq_retrieve

Execute BigQuery SQL and enrich rows into ACF-ready structured context.

Attribute	Value
Module(s)	retrievers/bq_retriever.py, bq_context_enrich.py, bq_ranking_cache.py, bq_trend_companion.py
Reads state	query, decomposition, bq_sql_plan, bq_table_candidates, structured_ranking_cache, bq_top_k, time overrides
Writes state	bq_results, bq_sql_queries, bq_sql_debug, structured_ranking_cache, bq_cache_hit
LLM	Yes (NL2SQL; templates without LLM when matched)
External I/O	BigQuery, optional Redis (schema cache)
Failure behavior	Diagnostic rows in bq_results; validation/execution errors in metadata.

Sub-steps

- Check ranking cache hit (list/export follow-up) — skip BQ if compatible.
- Template SQL fallback then NL2SQL per table hint.
- Validate SQL (SELECT-only, staging_dev allowlist, LIMIT).
- Execute against BigQuery; project_bq_row_acf on rows.
- enrich_bq_results: semantics prose, ranked_table consolidation.
- Stamp ACF metadata; FAOSTAT trend companion (Y-1 vs Y+1).
- trim_bq_result_contents; save ranking cache entry.

Node: merge

Combine BQ and vector chunks into unified merged_context with source labels.

Attribute	Value
Module(s)	chatbot/graph.py
Reads state	bq_results, vector_* lists, query
Writes state	merged_context
LLM	No

Attribute	Value
External I/O	None
Failure behavior	Empty merge if all sources empty.

Sub-steps

- Label BQ as [Structured data]; news/academic/policy/etc. with kind prefixes.
- Set `_context_kind` on each item (bigquery, news, academic, ...).
- Pin BQ first for ranking/numeric queries.
- Filter unusable diagnostic BQ rows.

Node: rerank

Score and trim merged `_context` to `reranked_context` using cross-encoder or API rerankers.

Attribute	Value
Module(s)	chatbot/reranker.py
Reads state	query, merged_context, decomposition, rerank_top_k
Writes state	reranked_context
LLM	Only when RAG_RERANKER_MODE=llm
External I/O	OpenRouter/Cohere APIs when configured
Failure behavior	Graceful degradation to cross_encoder then off.

Sub-steps

- Select mode: cross_encoder, openrouter, cohere, llm, off.
- Apply static source boosts (BQ ranked_table, numeric/comparative queries).
- Min-max normalize scores; combine with boost.
- Trim to rerank_top_k (default 20).

Node: web_fallback

Append supplemental Wikipedia/Tavily chunks when internal context is thin.

Attribute	Value
Module(s)	retrievers/web_retriever.py
Reads state	query, decomposition, reranked_context
Writes state	web_results, reranked_context (appended), web_fallback_status, web_fallback_reason, insufficient_context
LLM	No
External I/O	Wikipedia REST, Tavily API
Failure behavior	Fail-soft; may route to insufficient_context.

Sub-steps

- Trigger when usable chunks below RAG_WEB_FALLBACK_MIN_CHUNKS.
- Tier 1: Wikipedia shaped queries (no API key).
- Tier 2: Tavily if key set and daily limit not exceeded.
- Append web_wikipedia / web_search chunks to reranked_context.
- Set insufficient_context when web fails and context still thin.

Node: insufficient_context

Return deterministic gap message without LLM hallucination.

Attribute	Value
Module(s)	chatbot/graph.py, answer_language.py
Reads state	query, answer_lang
Writes state	answer, citations=[], acf=no_evidence
LLM	No
External I/O	None
Failure behavior	N/A (terminal node).

Sub-steps

- Build localized insufficient_context_answer.
- Set ACF band no_evidence.

Node: generate

Pack context, call LLM, extract citations, score ACF Path B on cited evidence.

Attribute	Value
Module(s)	chatbot/generator.py, acf_scoring.py, acf_metadata.py
Reads state	query, reranked_context, decomposition, conversation_summary, recent_turns, plan_type, category, answer_lang
Writes state	answer, citations, usage, acf_* fields, error
LLM	Yes (RAG_LLM_MODEL_ID / plan-specific model)
External I/O	LLM provider
Failure behavior	Context-only fallback message; error string on pipeline failure.

Sub-steps

- Filter/drop geo-conflicting context when configured.
- Build numbered [Source N] context block with char budgets.
- Apply plan/category/structured-BQ prompt guards.
- LLM generate; strip preamble and invalid [N] markers.
- referenced_citations + score_cited_evidence.

Node: export

Produce downloadable CSV/chart/DOCX/PDF when route enables exports and query requests them.

Attribute	Value
Module(s)	chatbot/export_runner.py, chatbot/exports/*
Reads state	export_enabled, export_intent, answer, citations, bq_results, reranked_context
Writes state	artifacts
LLM	Optional (report prose sections)
External I/O	GCS (production), local filesystem (dev)
Failure behavior	Empty artifacts[]; answer still returned.

Sub-steps

- Skip if export_enabled false or no export_intent.
- run_exports: tabular rows from BQ/ranked_rows, chart PNG, docx/pdf builders.
- Upload to GCS or local URI; attach ArtifactItem list.

Node: generate_meta

Answer identity/meta questions about Ask ADZA without retrieval.

Attribute	Value
Module(s)	chatbot/assistant_identity.py
Reads state	query, answer_lang
Writes state	answer, citations=[], curated ACF
LLM	No
External I/O	None
Failure behavior	N/A (terminal).

Sub-steps

- Return canned assistant identity response.

Node: generate_product

Answer product/how-to questions from curated knowledge.

Attribute	Value
Module(s)	chatbot/product_knowledge.py
Reads state	query
Writes state	answer, citations=[], curated ACF
LLM	No
External I/O	None
Failure behavior	N/A (terminal).

Sub-steps

- Match product FAQ patterns; no vector/BQ retrieval.

Node: generate_social

Handle greetings and out-of-scope queries politely.

Attribute	Value
Module(s)	chatbot/query_gate.py
Reads state	query, is_greeting_query, is_out_of_scope_query
Writes state	answer, citations=[]
LLM	No
External I/O	None
Failure behavior	N/A (terminal).

Sub-steps

- classify_social_query or generate_social_answer.

Node: generate_language_help

Ask user to rephrase when language cannot be detected.

Attribute	Value
Module(s)	chatbot/answer_language.py
Reads state	query, answer_lang
Writes state	answer, citations=[], curated ACF
LLM	No
External I/O	None
Failure behavior	N/A (terminal).

Sub-steps

- language_unclear_answer stub.

RAGGraphState field catalog

Field	Type	Description
query	str	Working question (may be enriched from memory)
enriched_query	str null	Set when query_enricher merged a follow-up
decomposition	dict	intent, entities, geography, domains, time_*, africa_default, africa_panel
measure_id	str null	Resolved agri measure ontology id
recency_tier	str null	live near_term historical_ok point_in_time
task_mode	str null	clarify analytical fact_lookup briefing data_export_only research chat
analytical_mode	bool null	True when task_mode == analytical
bq_table_candidates	list	Staging YAML table hints from bq_reason
bq_sql_plan	dict	selected_tables, query_intents, skip_bq, table_hints, measure_id
vector_news_results	list	News corpus chunks (VECTOR LEG)
vector_academic_papers_results	list	Academic papers chunks
vector_policies_results	list	Policy document chunks
vector_public_reports_results	list	Public report chunks
vector_formation_results	list	Formation corpus chunks
vector_ota_results	list	OTA insights chunks
vector_academic_results	list	Deprecated alias (news+academic convenience)
vector_results	list	Combined vector convenience list
corpus_selection	dict null	Corpus router output (selected keys, boosts)
bq_results	list	Enriched BigQuery context items (BQ LEG)
bq_sql_queries	list	Distinct SQL strings executed
bq_sql_debug	list	Per-query debug metadata
structured_ranking_cache	dict null	Session-scoped ranking cache entry
bq_cache_hit	bool null	True when ranking follow-up reused cache
merged_context	list	Vector + BQ (+ later web) before rerank
reranked_context	list	Subset passed to generator
web_results	list	Wikipedia/Tavily chunks from web_fallback
web_fallback_status	str null	ok disabled rate_limited empty error
web_fallback_reason	str null	Human-readable web fallback reason
insufficient_context	bool null	Route to gap answer after web failure
answer	str	Final assistant prose
citations	list	Structured citation objects
usage	dict	LLM token totals
error	str null	Pipeline-level error string
geo_override	str null	API/UI geo override
time_start_override	str null	Retrieval date filter override
time_end_override	str null	Retrieval date filter override
news_top_k	int null	Max news chunks

Field	Type	Description
academic_top_k	int null	Max academic chunks
ota_top_k	int null	Max OTA chunks
bq_top_k	int null	Max BQ rows/chunks
rerank_top_k	int null	Chunks after rerank
conversation_summary	str null	Rolling chat summary
recent_turns	list null	Last N verbatim Q/A pairs
chat_history	list null	Legacy client-owned history
is_meta_query	bool null	Route to generate_meta
is_product_query	bool null	Route to generate_product
is_greeting_query	bool null	Route to generate_social
is_out_of_scope_query	bool null	Route to generate_social
is_language_unknown	bool null	Route to generate_language_help
plan_type	str null	Free Farmers Government NGOs Agribusinesses Integrated
category	str null	Generation persona
user_profile	dict null	country, plan_type, category
acf_band	str null	ACF confidence band
acf_band_label	str null	Human-readable band
acf_score	int float null	0-100 composite score
acf_explanation	str null	One-sentence rationale
acf_components	dict null	Optional T/A/F/G breakdown
answer_lang	str null	Detected answer language tag
session_id	str null	Server session id
export_enabled	bool null	Allow artifact generation
export_intent	str null	csv chart docx pdf multi detected in query
artifacts	list	Downloadable export descriptors

Retrieval subsystems

Corpus router

- select_corpora (chatbot/corpus_catalog.py) chooses which of six collections to query.
- Inputs: query text, decomposition facets, plan_type, category, task_mode (briefing/research nudge news/academic).
- Heuristics + keyword/domain cues; stamps corpus_boost for merge/rerank.
- RAG_CORPUS_ROUTER=off restores fan-out to all six; never skips more than three corpora; never returns an empty set.
- Optional legacy mixed research collection only when RAG_USE_LEGACY_RESEARCH_COLLECTION=on.

Six corpora

Key	Qdrant collection	Role	Key payload indexes
news	news_data	Timely journalism	geo_country_primary, published_at, domains
academic_papers	academic_papers	Peer-reviewed research	country, publication_year
policies	policies	Policy documents	country, publication_year
public_reports	public_reports	Institutional reports	country, publication_year
formation	formation	Training / farmer practice	country, publication_year
ota	OTA_insights	Analyst insights (triple vectors)	country, insight_lanes

Vector retrieval (first-class peer to BQ)

- VECTOR LEG runs in parallel_retrieve BEFORE BQ reason/retrieve; both legs fuse at merge.
- Embeddings: fastembed (Railway) or sentence-transformers (local); E5 prefixes query: at search, passage: at index.
- Per corpus VectorRetriever: embed → payload filters (doc_kind, geo, time, news domains) → search.
- vector_search_mode: dense_named (single dense) or hybrid (dense + BM25 sparse RRF when RAG_QDRANT_HYBRID_SEARCH=on).
- Cascade (_retrieve_vector_cascade): full filters → time ±1 year → drop time → drop geo → drop both when env fallbacks allow.
- Surviving hits stamp constraint_relaxed when filters were widened; geo post-filter runs on all corpora after cascade.
- Soft-fail: empty corpus returns []; thread pool continues other corpora; graph never aborts on one empty collection.
- State writes: vector_news_results, vector_academic_papers_results, vector_policies_results, vector_public_reports_results, vector_formation_results, vector_ota_results, vector_results, corpus_selection.

BigQuery retrieval

- BQ LEG is peer to vector — not a replacement for unstructured retrieval.
- agri_measure_ontology.resolve_measure scopes candidate tables and filter hints (Yield vs Production, trade, FEWS prices, IPC, soil, climate).
- reason_bq_sql_plan: ontology-scoped YAML index → LLM with retries → ontology fallback_plan last resort (not silent skip when measure known).
- Forced plans for analytical / fact_lookup / data_export_only from MeasureSpec (investor_best_country multi-table).
- Modes: RAG_BQ_NL2SQL_MODE=per_hint (default) or batch; templates + patterns (element=Yield when yield).
- Validation: bq_sql_validate — SELECT-only, staging_dev tables, LIMIT enforced.

- Row budget: RAG_BQ_ROWS_PER_QUERY per SQL, capped by bq_top_k total; enrich + ranking cache + trend companion.

Ingest architecture (offline)

- Google Drive sync (ingestion/gdrive) or local files → text_processors/preprocess.
- JSONL under ml-eng/data/local/preprocessed_data/ (news, research, ota).
- Chunk IDs: UUID5 from chunking_config.CHUNK_ID_NAMESPACE.
- Loaders upsert to Qdrant; ingest_manifest.json tracks content_hash for incremental skip.
- Collection specs: scripts/qdrant_collection_specs.py (vectors, payload indexes).
- Rebuild: bump INGEST_VERSION or loader --reset after embedding/chunk changes.

LLM usage matrix

Step	Module	Backend	Notes
Decompose (optional)	query_decomposer.py	llm_chat	Heuristics always; LLM enriches when configured
BQ reasoner	bq_sql_reasoner.py	llm_chat	Ontology-scoped; retries; ontology fallback last resort
NL-to-SQL	bq_retriever.py	llm_chat	Templates/patterns first when matched
Rerank	reranker.py	cross_encoder default	LLM rerank only when RAG_RERANKER_MODE=llm
Generate	generator.py	llm_chat	Vector+BQ context; persona/recency overlays
Memory summary	chat_memory.py	llm_chat	Text stub if no LLM
Export prose	export_runner.py	llm_chat	Optional report sections

Observability and debugging

- Langfuse: rag_trace_context wraps API/chat turns; observed_span per graph node.
- POST /feedback on RAG API records trace scores via langfuse_trace_id.
- include_trace on POST /query returns decomposition + retrieval counts.
- Streamlit inspector: full state, bq_sql_debug, bq_cache_hit, ACF fields.
- Redis keys: rag:session:<session_id> (TTL RAG_SESSION_TTL_SECONDS, default 24h).

Environment variables (subset)

Variable	Purpose
BQ_PROJECT	BigQuery GCP project
BQ_DATASET_SILVER	Staging dataset for RAG SQL (default staging_dev)
QDRANT_URL / QDRANT_API_KEY	Vector database cluster
QDRANT_COLLECTION_NEWS	news_data override
QDRANT_COLLECTION_ACADEMIC_PAPERS	academic_papers override
QDRANT_COLLECTION_POLICIES	policies override
QDRANT_COLLECTION_PUBLIC_REPORTS	public_reports override
QDRANT_COLLECTION_FORMATION	formation override
QDRANT_COLLECTION_OTA_INSIGHTS	OTA_insights override

Variable	Purpose
RAG_CORPUS_ROUTER	Heuristic gate over six corpora (on default)
RAG_QDRANT_HYBRID_SEARCH	Dense + BM25 sparse RRF fusion
RAG_*_GEO_FALLBACK / RAG_*_TIME_FALLBACK	Per-corpus cascade relax flags
RAG_USE_LEGACY_RESEARCH_COLLECTION	Optional legacy mixed research collection
RAG_LLM_BASE_URL / RAG_LLM_API_KEY	OpenAI-compatible LLM backend
RAG_LLM_MODEL_ID	Default generation model
RAG_BQ_REASONER_MODEL_ID	Dedicated BQ table-reasoner model
RAG_BQ_REASONER_TIMEOUT_S	Reasoner HTTP timeout override
RAG_BQ_REASONER_RETRIES	Reasoner retry count (default 3)
RAG_BQ_NL2SQL_MODEL_ID	Dedicated NL-to-SQL model
RAG_RERANKER_MODE	cross_encoder openrouter cohere llm off
RAG_RERANKER_MODEL	Cross-encoder model (default BAAI/bge-reranker-base)
RAG_REDIS_URL	Session blobs + shared caches
RAG_SESSION_TTL_SECONDS	Session TTL (default 86400)
RAG_WEB_FALLBACK_ENABLED	Post-rerank Wikipedia/Tavily (default off)
RAG_CITATIONS_MODE	referenced (default) or all
RAG_GENERATE_MAX_TOKENS	Answer length cap (default 2048)
RAG_GENERATE_CONTEXT_MAX_CHARS	Total context chars (default 12000)
LANGFUSE_*	Tracing and POST /feedback scores

Related documentation

- [OpenTrace-RAG-Pipeline-Architecture.docx](#) — Word version with ASCII diagrams
- [OpenTrace-RAG-API-Documentation.docx](#) — HTTP /query reference
- [OpenTrace-Chatbot-API-v1-Documentation.docx](#) — public v1 chat + artifacts
- [ml/rag/docs/API.md](#) — markdown API reference
- [ml/rag/ARCHITECTURE.md](#) — package architecture (markdown)
- Regenerate PDF: `python scripts/generate_rag_architecture_pdf.py` from `ml-eng/`